# 3.05 Train Test Split / Cross-Validation

Katie Sylvia

GENERAL ASSEMBLY

# Learning Objectives

- Describe train/test split and cross-validation.
- Explain how these validation techniques differ and why we want to use them.
- Split data into testing and training sets using both train/test split and cross-validation and apply both techniques to score a model.

# Review Bias & Variance

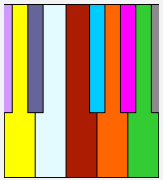
# This is what we learned in class:

## Error #1: Bias



***Bias*** **is how bad your model is at predicting *y*.**

## Error #2: Variance



***Variance*** **is how bad your model is at generalizing to new data.**



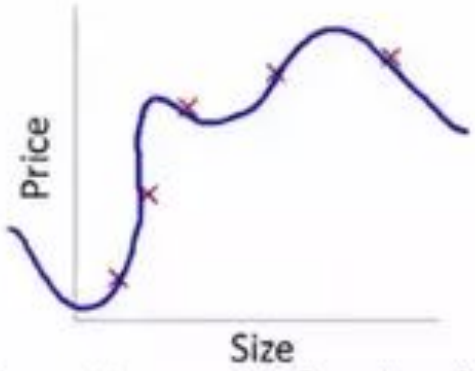

$\theta_0 + \theta_1 x$

High bias
(underfit)




$\theta_0 + \theta_1 x + \theta_2 x^2$

"Just right"




$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

High variance
(overfit)

# Why does bias occur?

- Our model is too simple for our data.
- On average, our model $\hat{y}$ is going to be far from the truth $y$.
    - Ex. I tried to model a curved relationship with a straight line.
- When we rely on simplifying assumptions that aren't valid (i.e.linearity), we can run into high bias.
- Linear regression is a method where we might suffer from high bias.

# Why does variance occur?

- Basically, more variance points us to believe our model is too complex.
- Our model $\hat{y}$ matches our data too closely.
- May not perform well on data it hasn't seen yet.
- We may not have enough data.
- Our model may "fit" very well to data it's seen, but not generalize well to data it hasn't.

# How to Handle Bias and Variance



## What can we do?

**Strategies**

- Get more data
- Get a better sample of data
- Increase complexity of model
- Decrease complexity of model
- Regularize
- Later: ensemble methods

# Data Science at its core:

- Build a model
- Make predictions

Key assumption: there are 2 datasets out there: one that our model trains on (training set), and one that we make make predictions on (test set). This is important in order to see how our model performs on unseen data!

In order to truly evaluate your model you need two scores:

- A score that the model was trained on
- A score from data that the model was not trained on.

Bias-Variance Tradeoff

## Bias

**Complexity?**

LOW

**Fit?**

Underfit

**Accuracy (on train)?**

LOW

## Variance

**Complexity?**

HIGH

**Fit?**

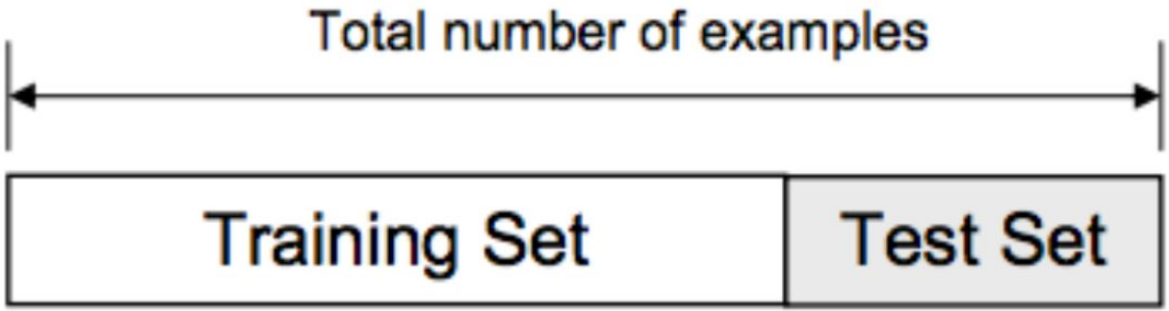
Overfit

**Accuracy (on train)?**

HIGH

*Note: these are *general rules*






Thankfully, sklearn has a great method to split our data for us. The default is have 25% of the data reserved for the test set.
Let's go see this in action!



# How do we diagnose high/low bias/variance?

# Train Test Split






$\theta_0 + \theta_1 x$

High bias
(underfit)

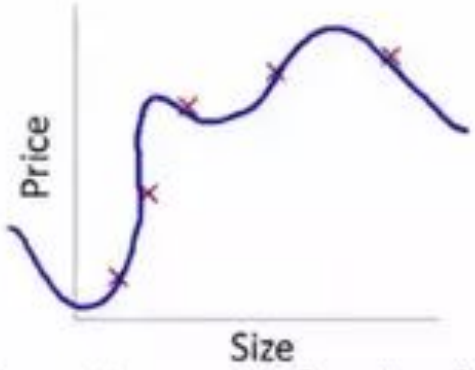
Train/Test scores are **close to the baseline score**




$\theta_0 + \theta_1 x + \theta_2 x^2$

"Just right"

Train/Test scores are **fairly similar** and reasonably (*subjective!) **higher than the baseline score**




$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$

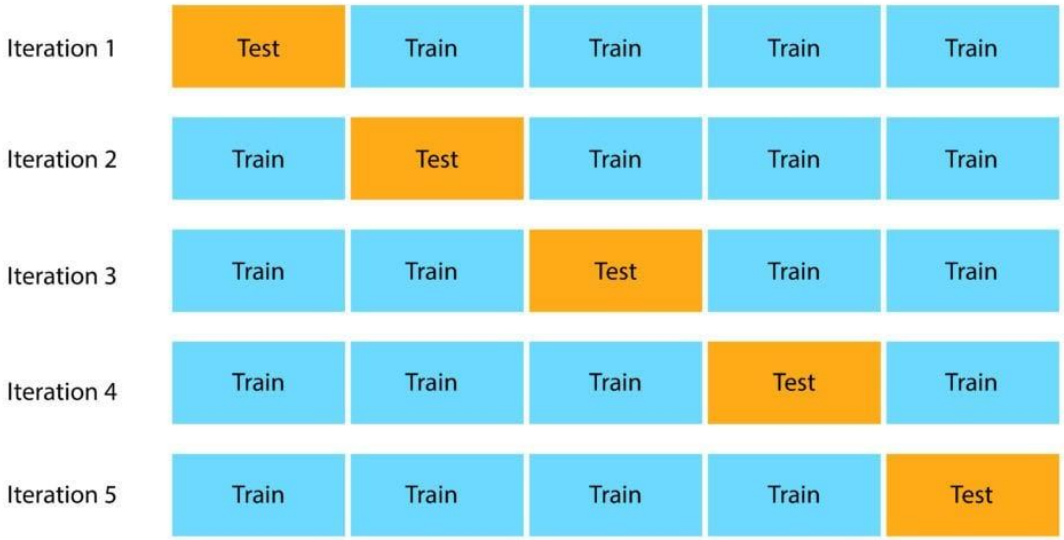
High variance
(overfit)

Train/Test scores are **far apart**

# K-Folds Cross Validation



K-fold cross-validation takes the idea of a single train/test split and expands it to multiple tests across different train/test splits of your data.




Default is 5 folds.


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